

Privacy-Preserving Data Mining

- Privacy leakage between users and data owners
 - Example: re-identify users based on anonymous reviews on products.
- Anonymize data for privacy protection
 - k-anonymity (P. Samarath et al., 1998): generalize the data so that it is identical to at least $k-1$ other records on the identifying attribute.
- Differential privacy (A. Machanavajjhala et al. 2006)
 - Describe patterns of groups without sharing individual information.
 - Global sensitivity: for a function f and a pair of neighboring datasets D_1 and D_2 , GS_f is the maximum difference between function value $f(D_1)$ and $f(D_2)$.

Privacy-Preserving Data Mining

- ❑ Privacy leakage between data owners
 - ❑ Example: multiple companies collaboratively conduct data mining but refuse to share user information.
 - ❑ Solution: federated learning and analytics.
- ❑ Federated learning
 - ❑ Run local computation over each data owners' devices and aggregate results without leaking detailed data from individual data owners.
 - ❑ Example: federated K-Means.
 - ❑ Horizontal and vertical federated learning
 - ❑ Horizontal: data owned by each data owners follows the same schema.
 - ❑ Vertical: different data owners have the data on only some attributes/objects.